

SAMAR ANAND

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EDUCATION

VIT Bhopal University

B.Tech in Computer Science (Artificial Intelligence and Machine Learning)

CGPA: 8.63/10.00

Bhopal, Madhya Pradesh

Aug 2023 – Aug 2027

TECHNICAL SKILLS

- **Programming:** C++, Java, Python, JavaScript, TypeScript, SQL
- **Web Technologies:** ReactJS, Next.js, Node.js, Express.js, FastAPI, REST APIs, Socket.io, WebSockets, HTML, CSS, Tailwind CSS, Material UI
- **Databases & Tools:** MongoDB, PostgreSQL, MySQL, Git, GitHub, Docker, Microsoft Azure, Vercel, Render
- **AI & Machine Learning:** NLP, TF-IDF, Scikit-learn, RAG, LangChain, LangGraph, Hugging Face

PROJECTS

VibeCode — AI-Powered Web IDE | *Next.js 15, TypeScript, WebContainers, Monaco Editor, Gemini API, Prisma, MongoDB* | [GitHub](#)

- Built a browser-based AI-integrated IDE using Next.js 15 and WebContainers, supporting code compilation and execution across multiple frameworks including React, Next.js, Express, Hono, Vue, and Angular.
- Integrated Google Gemini API (gemini-2.5-flash) with a local Ollama fallback for AI-powered code completion and context-aware project refactoring assistance.
- Implemented Edge-compatible NextAuth v5 authentication with Google and GitHub OAuth, along with Prisma and MongoDB for persistent workspace and project data.
- Integrated an interactive xterm.js terminal with WebContainer system shells and a custom file explorer, enabling zero-setup workspace creation and file CRUD operations within the browser.

MedQueue AI — Real-Time Triage & Priority Queue | *React 18, Node.js, Express, Socket.io, Gemini API, Zod, MongoDB* | [GitHub](#)

- Architected a real-time clinical intake platform using React 18, Node.js, Express, Socket.io, and MongoDB to automate patient intake and triage queueing.
- Integrated Google Gemini API (gemini-2.5-flash) with Zod structured output parsing and a zero-downtime local rule fallback engine to classify Emergency Severity Index (ESI 1–5) tiers and detect life-threatening red flags.
- Engineered an Atomic Queue Re-Ordering engine with ACID transaction locks, allowing high-risk ESI 1–2 emergency patients to automatically jump queue ranks in real time.
- Streamed live bi-directional WebSockets via Socket.io across concurrent dashboards (Nurse Station, Doctor Workbench, ER Lobby Board), deploying frontend on Vercel and backend on Render.

Nexus Cart — Voice-Powered Shopping Assistant | *React, Node.js, Express, MongoDB, Gemini API, NLP* | [GitHub](#)

- Built a voice-driven e-commerce assistant using React, Node.js, and the Web Speech API for voice-based product interaction and real-time shopping state synchronization.
- Integrated Google Gemini API for NLP-based command processing, achieving high classification accuracy across item categorization, unit conversion, and localized price estimation tasks.
- Optimized MongoDB Atlas data operations using Mongoose upsert logic, improving database query efficiency while preventing duplicate shopping records.
- Developed an automated inventory restock scheduling system with exponential backoff and retry handling for API rate-limit constraints.

EXPERIENCE

Ethnus

May 2025 – Jul 2025

MERN Stack Intern

Remote

- Developed RESTful backend services for core platform modules.
- Optimized MongoDB schemas and queries, improving data retrieval speed and database efficiency.
- Developed responsive React interfaces across key application pages, enhancing user experience and rendering performance.
- Implemented authentication and role-based access control (RBAC) to secure application endpoints and user roles.

ServiceNow

May 2026 – Jun 2026

Virtual Internship Program — ServiceNow University, AICTE & SmartBridge

Remote

- Worked with ServiceNow administration concepts including user management, platform configuration, workflow automation, reporting, and dashboard creation.
- Explored Agentic AI capabilities and automated workflow development within the ServiceNow platform.
- Tested custom application logic using ServiceNow Automated Test Framework (ATF) configurations.

CERTIFICATIONS & ACHIEVEMENTS

- **AlgoUniversity Programming Camp:** Selected as a mentee among 60,000+ participants; featured in the Hall of Fame.
- **Leadership:** Selected as an Open Source Contributor for GirlScript Summer of Code (GSSoC) and appointed as Campus Ambassador for Internsala student program and GSSoC.
- **Certifications:** Microsoft Azure Fundamentals DP-900(Microsoft), MERN Full Stack Development (Ethnus), Applied Machine Learning in Python (University of Michigan).